

Superstore Sales & Profit Analysis

End-to-end data analytics project • Python • MySQL • Power BI

Overview

This project analyzes the Superstore retail dataset to evaluate sales, profitability, product performance, customer segments, returns, and regional performance. The workflow combines Python-based analysis, relational SQL analysis, and an interactive Power BI dashboard to convert transactional data into business-focused insights.

Objectives

- Evaluate overall sales, profit, order volume, average order value, and profit margin.
- Compare performance across regions, categories, sub-categories, and customer segments.
- Identify the highest- and lowest-performing products.
- Analyze returned orders and regional manager performance.
- Examine sales and profit trends over time.
- Build an interactive dashboard suitable for business reporting.

Technology Stack

Technology	Application
Python / Pandas	Data cleaning, preparation, EDA
Matplotlib	Exploratory data visualization
MySQL	SQL analysis, aggregation and joins
Power BI	Data modeling, DAX, interactive dashboard
GitHub	Project documentation and version control

Dataset

The analysis uses three related Superstore tables:

Orders — Transactional sales data including dates, customers, geography, products, sales, quantity, discount and profit.

Returns — Order-level return information linked through Order ID.

People — Regional manager information linked through Region.

Data Preparation & Analysis

- Inspected the dataset for missing values and duplicate records.
- Validated data types and standardized fields required for analysis.
- Prepared Orders, Returns, and People tables for relational analysis.
- Used MySQL to aggregate metrics and join related tables.
- Created calculated metrics required for dashboard reporting.

Power BI Report

The Power BI report is structured into two pages with distinct analytical purposes.

01 | Executive Overview

KPI cards: Total Sales, Total Profit, Total Orders, Average Order Value, Profit Margin.

Visuals: Sales & Profit by Region, Category, Sub-Category and Segment, followed by a Sales & Profit Trend.

Purpose: Provide a concise view of overall business performance and major trends.

02 | Product & Regional Analysis

Filters: Order Year, Region, Category and Segment.

Visuals: Profit by Sub-Category, Top 10 Products by Profit, Bottom 10 Products by Profit, Returned vs Not Returned Orders, and Sales & Profit by Regional Manager.

Purpose: Identify product opportunities, loss-making products, return patterns and regional performance.

Data Model

The analytical model connects the transactional Orders table with Returns through **Order ID** and with People through **Region**. This structure enables product, return, geographic and manager-level analysis.

Key Business Questions

- Which regions contribute the most sales and profit?
- Which categories and sub-categories are most profitable?
- Which products are the strongest and weakest performers?
- Which customer segment contributes the most business?
- What proportion of orders are returned?
- How do regional managers compare on sales and profit?
- How do sales and profit change over time?

Project Structure

```
Superstore-Sales-Analysis/  
■■■ data/  
■■■ notebooks/  
■■■ sql/  
■■■ powerbi/  
■■■ images/  
■■■ README.md
```

Skills Demonstrated

Data Cleaning • Exploratory Data Analysis • Python • Pandas • SQL • MySQL • Data Modeling • Table Joins • DAX • Power BI • Dashboard Design • Business Analysis • Data Storytelling

Future Enhancements

- Add year-over-year growth and variance metrics.
- Introduce customer-level segmentation and retention analysis.
- Add dedicated return-rate and return-impact analysis.
- Implement Power BI drill-through pages and advanced DAX measures.

- Enable automated refresh and online dashboard deployment.

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This README is designed as professional project documentation for portfolio and GitHub use.